

Glider Selection Dashboard

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Decision Support System

Glider Selection Dashboard — Match requirements to configurations

- Slack channel: # glider-selection-dashboard
- GitHub repo: /glider-selection-dashboard

1. Define Mission Constraints

Users input environmental profiles and project parameters:

- Mission duration, geographic area & distance/area to cover
- Bathymetry, operating depth & environmental conditions
- Species/phenomena of interest & required sensors
- Sampling rate, duty cycle & real-time vs. archival
- Operational constraints, endurance, battery type & budget

2. Intelligent Suitability Assessment

The system evaluates and optimizes platform configurations:

- Generates a ranked shortlist with distinct fit justifications
- Side-by-side platform comparison & trade-off analysis
- Pre-deployment risk identification for battery & telemetry limits

- Potential Data Sources/Tools: Manufacturer websites
- Potential Challenges: ??
- First steps: Where to create the Dashboard? R (shinydashboard)?

AI inspiration



Inputs → shortlist → comparison → mission feasibility

Dynamic Dashboard w/ Info-

- Info from manufacturers

Work with Quarto and obsJS??

Where to get the info to build this?? What kind of info do you need besides the sensor data?
Important thing to consider is power consumption profile

There is this website to consider: <https://github.com/smerckel/gliderflight>

Important to have very detailed power consumption

- Slocum has this info so you can track it in real-time
- Seaglider - only newer models track this at that level of detail

GOAL is to create the template (put in some placeholder numbers) and then work with the community to improve the accuracy of the input data, etc.

Does a tool exist that helps you plan a survey? If you have a particular kind of glider that you have to work with (rather than selecting)

- Slocum has a "simulator" but it isn't great
- SeaExplorer does have a planning tool in their piloting tool GLIMPSE
- Nothing that we know of exists for Seaglider

- This could be an add on to this project - “Ok, I’m going to go with this glider, in this area, what can I accomplish?”

One useful feature of the tool could be to highlight unique affordances/abilities of each glider

Quarto file (.qmd) are edited in RStudio

-can include code chunks in R or Python

-see Info on [Observable JS](#) in hackathon resources

Heloise could not make the breakout session, asked Anne/Ehsan to identify what skills they bring to the table, and what they’d like to learn/get out of this....

- Anne wants to learn more about the different gliders & their capabilities, she has experience with quarto & happy to help with that side of things.
- Ehsan- exact opposite! No experience w/ quarto, but experience with other types of gliders. Experience with C, C++, some Python
- Kasey - no experience with gliders, or quarto. Good coding and PAM experience. Mostly hoping to learn from the group

Ehsan shares [Glider Operation Best Practices](#) (including mission planning) and [Glider PAM considerations](#)

Dashboard Components - [see spreadsheet w/glider comparisons](#)

1. Yes/No questions with level of importance

Do we get anything back in real-time?

What can pilot change during the mission?

- Sampling rate - frequency
- Depth
- Duty cycle

Can you make the glider be at a specific depth at a specific time?

Can you fly the glider using a bathymetry file

Are there existing edge-computing technologies for on-board processing?

-<https://github.com/e-abdi/SMaRT>

-Seaglider and Slocum can’t change internal routines; but “back-seat driver” options exist

Logistics

Ease of deployment/recovery depending on the vessel availability

-requirement of crane/basket for recovery vs. small boat

Batteries

Rechargeables or Primay

Notes on communication/business model of manufacturer

Altimeter

Seaglider ~12KHz

Slocum - 170KHz

Seaexplorer

OceanScout

Spray

2. Endurance Calculations (power and storage)